

Subject Code	AF5641
Subject Title	Econometrics for Data Analysis
Credit Value	3
Level	5
Normal Duration	One Semester
Pre-requisite / Co-requisite/ Exclusion	Nil
Objectives	This subject contributes to Learning Outcome 1 by equipping students with the knowledge and skills to understand and apply econometric methods in various fields. This enhances their comprehensive understanding of modern economic theories and applications. It contributes to Learning Outcome 2 by developing students' skills in data science and the digital economy. It does so by introducing students to econometrics and its role in data analysis, exploring basic statistical concepts in probability theory and hypothesis testing, and delving into linear regression. Moreover, this subject significantly contributes to Learning Outcome 3 by equipping students with the necessary analytical and technical skills to perform a wider range of real-life analysis. The second half of the course introduces limited dependent variable models, panel data models, and time series analysis, providing students with a comprehensive introduction to some of the most important areas of modern econometrics. This practical application of knowledge and skills ensures that students are well-prepared to tackle real-world problems in business, economics, finance, and policy.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - Collect, clean and prepare raw data for econometric analysis. - Read, interpret and evaluate fundamental econometrics theories and arguments used in economics literature. - Apply appropriate data analysis and econometrics methods to real-world datasets. - Understand common statistical problems encountered in research and determine suitable econometric techniques for various circumstances. - Integrate knowledge from related subjects in research design. - Utilize statistical software (i.e. Stata, R) to implement and interpret the econometric methods covered in the course.

Subject Synopsis/ Indicative Syllabus	• Introduction to econometrics: its role in data analysis; Probability theory, Hypothesis testing and confidence intervals; • Linear Regression: Model Specification and Diagnostic Testing: Understanding the problem of imperfect multicollinearity and heteroscedasticity; Diagnostic testing for these issues; Remedial measures that can be taken; • Limited Dependent Variable Models: Logit and Probit models; • Panel Data Models; Fixed effects vs. random effects models • Time Series Analysis: Understanding and displaying time series data; Autoregressive and moving average models; the problem of spurious regressions; Other key issues related to time series.																																																																						
Teaching/Learning Methodology	A three-hour weekly seminar will be structured to guide and promote students’ understanding of relevant concepts. In addition, weekly homework’s, that involve applying the knowledge learnt to help build confidence with the material.																																																																						
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="6">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th><th>f</th></tr><tr><td>Attendance and participation</td><td>10</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Midterm test</td><td>20</td><td></td><td>✓</td><td></td><td>✓</td><td>✓</td><td></td></tr><tr><td>Final exam</td><td>50</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Individual and group assignments.</td><td>20</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td colspan="6"></td></tr></table> To reflect the technology content in this subject, 10% (or more) of the overall weighting of this subject is based on assessment concerning data analytics knowledge. Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: The assessment methods have been carefully chosen to evaluate the intended learning outcomes in a comprehensive manner. Continuous assessment will be employed to gauge your practical abilities, specifically in data handling, programming, and interpretation skills. This method allows for ongoing feedback and the opportunity to improve and refine your skills over time. On the other hand, the midterm and final exams are designed to assess your conceptual understanding of the course content.	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)						a	b	c	d	e	f	Attendance and participation	10	✓	✓	✓	✓	✓	✓	Midterm test	20		✓		✓	✓		Final exam	50	✓	✓	✓	✓	✓	✓	Individual and group assignments.	20	✓	✓	✓	✓	✓	✓																								
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	These exams will test your ability to recall, analyze, and apply the theoretical knowledge you have acquired throughout the course. This combination of continuous assessment and examinations ensures a balanced evaluation of both practical application and theoretical understanding, aligning with the intended learning outcomes.	
Student Study Effort Expected	Class contact:	
	Lectures / Seminars	39 Hrs.
	Other student study effort:	
	- Individual study - Assignments, Quizzes	36 Hrs. 36 Hrs.
	Total student study effort	111 Hrs.
Reading List and References	Textbooks: <i>Introductory Econometrics: A Modern Approach</i>, 6th Edition by Jeffrey Wooldridge. (You are welcome to use an older/newer version of the textbook/online pdf.) <i>Principles of Econometrics</i>, 5th Edition by R. Carter Hill, William E. Griffiths and Guay C. Lim. <i>Mostly Harmless Econometrics: An Empiricist's Companion</i> by Joshua Angrist and Jorn-Steffen Pischke. <i>An Introduction to R</i> by W.N.Venables, D.M.Smith, and the R Core Team <i>Introduction to Econometrics with R</i> by Christoph Hanck, Martin Arnold, Alexander Gerber, and Martin Schmelzer (Link: https://www.econometrics-with-r.org/) <i>Econometrics by Example</i> by Damodar Gujarati, D. Bloomsbury Publishing. <i>Introduction to Econometrics</i> by James H. Stock and Mark W. Watson. Pearson. Recommended further reading: <i>Mastering 'metrics: The Path from Cause to Effect</i> by Joshua D. Angrist and Jorn-Steffen Pischke. <i>Econometrics: Panel Data Methods</i> by Jeffery M. Wooldridge.	